Antonio Varagnolo

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Education

Georgia Institute of Technology, Atlanta, GA, USA — Fall 2023–present

Ph.D. in Computational Science and Engineering; *M.Sc. in Mathematics. Advisor:* Raphaël Pestourie. *Research interests:* scientific machine learning, AI-enhanced PDE-constrained optimization, transfer and meta-learning, stochastic and constrained optimization. *Coursework:* Deep Learning, Scientific ML, Algorithms and HPC; Analysis, Probability, PDEs, Numerical Analysis.

Bocconi University, Milan, Italy — Fall 2020–Summer 2023

B.Sc. in Economics and Computer Science. Thesis: "Stochastic differential equations: modeling and simulation in tumor immunology", advised by Prof. Giacomo Zanella.

Publications and Manuscripts

[1] **A. Varagnolo**, M. Dai, B. Deng, R. Pestourie. "Efficient Optimization of Mechanical Metamaterials via Transferable Representation Learning." In review, 2026.

[2] **A. Varagnolo**, Y. Pimonova, M. G. Taylor, R. Pestourie, N. Lubbers. "Interpretable Meta-Learning for Multi-Objective Chemical Search." In review, 2026.

[3] **A. Varagnolo**, A. Hammond, G. Romano, R. Pestourie, S. G. Johnson. "Adaptive Convex Approximations for Constrained Stochastic Optimization." In review, 2026.

[4] **A. Varagnolo**, G. Romano, R. Pestourie. "Physics Enhanced Deep Surrogates for the Phonon Boltzmann Transport Equations." *ASME Journal of Heat and Mass Transfer*, 2026.

Research and Work Experience

Graduate Research Assistant, Georgia Institute of Technology — Fall 2024–present

- **Transferable Inverse Design of Mechanical Metamaterials.** Transfers a latent representation learned on linear properties to nonlinear metamaterial design, experimentally validated. Reaches energy absorption within 5% of state-of-the-art optimization at ∼43% of the computational cost, with savings from the first design campaign.
- **Meta-learning for Multi-objective Molecular Discovery** (with Los Alamos National Laboratory). Interpretable linear meta-learning surrogate for multi-objective molecular discovery, run as a massively parallel pipeline on LANL HPC (600 CPUs). On organic-metal complexes it outperformed state-of-the-art methods (YY better hypervolume, XX better IGD+) while yielding chemical insight.
- **Constrained Stochastic Optimization.** Adaptive constrained stochastic optimizer with convergence guarantees (stochastic successive convex approximation). Converges up to 100× faster on classic problems, cuts worst-label MNIST error YY-fold, and sets the KL-reconstruction tradeoff when training VAEs.
- **Surrogate Modeling for Nanoscale Heat Transfer.** Surrogate combining a learned latent space with a low-fidelity solver for nanoscale thermal materials; cuts expensive Boltzmann transport solves by 70%.

Founding Scientist, Deleon Technologies — Fall 2024–Spring 2025

Built a digital twin of the human metabolome and an end-to-end microfluidic chip design pipeline; helped raise ∼$1M at an $8M valuation.

Research Assistant, Technical University of Munich (TUM) — Summer 2024

Adapted KANs and neural ODEs to thermoacoustic instabilities in combustion engines (advisor: Prof. Wolfgang Polifke).

Fellow, Tortuga Think Tank — Fall 2021–Summer 2023

Built the first Italian municipality-level House Affordability Index; a version ran in print in *La Repubblica*, a leading national daily.

Teaching and Course Development

Inverse Problems (CX/MATH 4741) — Spring 2027

Helped design topics, labs and exams.

Scientific Machine Learning (CSE 7752) — Fall 2026

Ran the student symposium.

Honors

Applied Machine Learning Fellow, Los Alamos National Laboratory — 2025

NSF Traineeship, Sustainable Development of Smart Medical Devices (SUSMED) — 2025